MediCare — Database Design & Data Modeling

DEPI Graduation Project

2. Database Design & Data Modeling

This document describes the relational database structure, entity relationships, normalization principles, integrity constraints, and physical table specifications for the MediCare Clinic Management System. The design below is aligned across the ERD, relationship matrix, and table definitions.

2.1 Entity-Relationship Diagram (ERD)

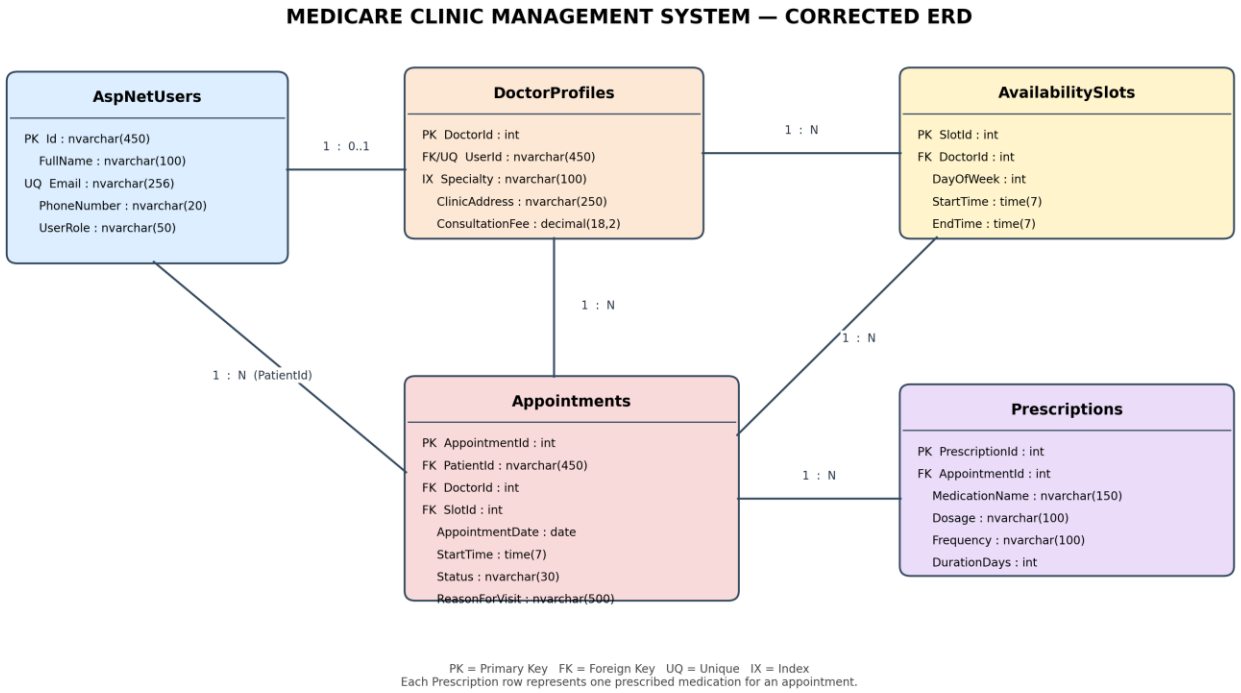


Figure 2.1: Corrected logical Entity-Relationship Diagram (ERD)

Entities and Relationships

- **AspNetUsers:** Stores authentication and shared account information for patients, doctors, and administrators. The UserRole value identifies the account role. Patient appointments reference this table directly.
- **DoctorProfiles:** Stores doctor-specific professional and clinic information. Each doctor profile belongs to exactly one application user; an application user may have zero or one doctor profile.
- **AvailabilitySlots:** Stores a doctor's recurring weekly availability using the day of week, start time, and end time. Each slot belongs to one doctor.
- **Appointments:** Stores each booked consultation, including the patient, doctor, selected availability slot, date, start time, status, and optional reason for visit.
- **Prescriptions:** Stores prescribed medication details. Each row represents one medication prescribed for an appointment; an appointment may therefore have zero or multiple prescription rows.

Relationship Matrix

Primary Entity	Related Entity	Cardinality	Relationship
AspNetUsers	DoctorProfiles	1 : 0..1	A user account may have one doctor profile; each doctor

DoctorProfiles	AvailabilitySlots	1 : N	profile belongs to one user. A doctor can define multiple recurring availability slots.
DoctorProfiles	Appointments	1 : N	A doctor can have multiple appointments; each appointment references one doctor.
AspNetUsers (Patient)	Appointments	1 : N	A patient can book multiple appointments; each appointment references one patient account.
AvailabilitySlots	Appointments	1 : N	A recurring availability slot can be referenced by multiple dated appointments.
Appointments	Prescriptions	1 : 0..N	An appointment may have no prescription rows or multiple medication rows.

2.2 Logical & Physical Schema

The database is implemented in Microsoft SQL Server and managed through Entity Framework Core Code-First Migrations. The schema is designed to support the stated entities and relationships while reducing avoidable duplication.

Normalization & Integrity Rules

- **First Normal Form (1NF):** Each column stores a single atomic value. For example, medication details are stored in individual prescription rows rather than as a delimited list in one field.
- **Second Normal Form (2NF):** Every non-key attribute depends on the whole primary key. The tables use single-column primary keys, so partial-key dependencies do not arise.
- **Third Normal Form (3NF):** Non-key attributes are stored with the entity they describe; avoid duplicating user or doctor profile attributes in appointment records.
- **Doctor double-booking prevention:** Create a composite UNIQUE index on (DoctorId, AppointmentDate, StartTime). Do not create separate unique indexes on AppointmentDate and StartTime, because that would incorrectly prevent different doctors or dates from using the same time.

Physical Database Table Specifications

The following table definitions are the authoritative physical schema represented in the corrected ERD. PK = Primary Key, FK = Foreign Key, UQ = Unique constraint, and IX = non-unique index.

Table 1: AspNetUsers (Identity Base User Table)

Column Name	Data Type	Keys / Indexes	Null?	Description
Id	nvarchar(450)	PK	No	Unique user identifier (ASP.NET Core Identity key).
FullName	nvarchar(100)	—	No	User's full display name.
Email	nvarchar(256)	UQ (recommended)	No	Account email; enforce uniqueness if used as the login identifier.
PhoneNumber	nvarchar(20)	—	Yes	Contact phone number.
UserRole	nvarchar(50)	IX (optional)	No	Account role: Patient, Doctor, or Admin.

Table 2: DoctorProfiles

Column Name	Data Type	Keys / Indexes	Null?	Description
DoctorId	int	PK, Identity	No	Primary key.
UserId	nvarchar(450)	FK, UQ	No	References AspNetUsers(Id); unique to enforce the one-to-zero-or-one relationship.
Specialty	nvarchar(100)	IX (recommended)	No	Medical specialty (e.g., Cardiology).
ClinicAddress	nvarchar(250)	—	No	Clinic physical location.
ConsultationFee	decimal(18,2)	—	No	Consultation price.

Table 3: AvailabilitySlots

Column Name	Data Type	Keys / Indexes	Null?	Description
SlotId	int	PK, Identity	No	Primary key.
DoctorId	int	FK	No	References DoctorProfiles(DoctorId).
DayOfWeek	int	CHECK: 0–6	No	Day of week (0 = Sunday, 6 = Saturday).
StartTime	time(7)	—	No	Availability start time.
EndTime	time(7)	—	No	Availability end time; must be later than StartTime for same-day slots.

Table 4: Appointments

Column Name	Data Type	Keys / Indexes	Null?	Description
AppointmentId	int	PK, Identity	No	Primary key.
PatientId	nvarchar(450)	FK	No	References AspNetUsers(Id) for the patient account.
DoctorId	int	FK	No	References DoctorProfiles(DoctorId).
SlotId	int	FK	No	References AvailabilitySlots(SlotId).
AppointmentDate	date	Part of composite UQ	No	Calendar date of the appointment.
StartTime	time(7)	Part of composite UQ	No	Booked start time; should be within the referenced slot's availability.
Status	nvarchar(30)	—	No	Pending, Confirmed, Cancelled, or Completed.
ReasonForVisit	nvarchar(500)	—	Yes	Optional patient notes or reason for visit.

Table 5: Prescriptions

Column Name	Data Type	Keys / Indexes	Null?	Description
PrescriptionId	int	PK, Identity	No	Primary key.
AppointmentId	int	FK, IX	No	References Appointments(AppointmentId). Multiple rows may belong to one appointment.
MedicationName	nvarchar(150)	—	No	Name of the prescribed medication.
Dosage	nvarchar(100)	—	No	Dosage instructions (e.g., 500 mg).
Frequency	nvarchar(100)	—	No	Frequency instructions (e.g., twice daily).
DurationDays	int	CHECK: > 0	No	Treatment duration in days; must be positive.

Implementation Notes

- The composite unique index for appointments must be defined as UNIQUE (DoctorId, AppointmentDate, StartTime). It should not be implemented as three separate unique indexes.
- DoctorProfiles.UserId must have a UNIQUE constraint in addition to its foreign key so that one user cannot own multiple doctor profiles.
- Appointments.PatientId references AspNetUsers.Id, and the application must validate that the referenced user has the Patient role.
- Appointments.DoctorId should match the doctor who owns the referenced AvailabilitySlots row. Enforce this consistency in application logic or by using an appropriate composite key/constraint design.
- The current schema does not define PatientProfiles, MedicalRecords, or ActivityLogs. They have therefore been excluded from the ERD and entity list rather than being shown as implemented tables.
- If the project later needs multiple clinical records per appointment, attached files, audit history, or a separate patient profile, those should be designed as additional tables with their own keys and relationships.